

OPIHI Data Overview

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marinelifephotograph
https://kamfamily.wordpress.com/2011/06/14/12-mouthwatering-tastes-of-maui/
https://seagrant.soest.hawaii.edu/coastal-and-climate-science-and-resilience/ccs-projects/what-is-a-king-tide/



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https://kamfamily.wordpress.com/2011/06/14/12-mouthwatering-tastes-of-maui/
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Hawaiian intertidal has been understudied

<u>Problem</u>:
Difficult to research



Modest tidal range (~1 m)



Seasonal waves

<u>Solution</u>:
Participatory Science

Public collection of scientific data







‘Opihi vs OPIHI

Our Project In Hawai‘i’s Intertidal

TWO OPIHI PROGRAMS

Participatory Science

K–12 teachers
(middle & high school)
& their students



BIOL 499
(dedicated section)
CURE

Undergraduates



TWO OPIHI PROGRAMS

Participatory Science

K–12 teachers
(middle & high school)
& their students



Scientific & educational goals

Stewardship & connection to **place**

Authentic **research experiences**

BIOL 499

(dedicated section)

Upper-level undergraduates



OPIHI Participatory Science Goals

Educational :

Engage students and teachers in place-based, field-based research, connect them to their environment, and enhance their scientific literacy skills.



Picture by K. Seraphin

Scientific :

Describe and monitor the Hawaiian rocky intertidal to determine if and how the community is changing over time.



Picture by N. Nahale-a & K.Swain

Rocky Intertidal



Picture by OPIHI



http://www.honolulumagazine.com/Honolulu-Magazine/November-2011/The-Opihi-Shellfish-Story/?cparticle=2

Photo: Josh Fletcher

INTERTIDAL:
Explore the Ocean





Pictures by N. Nahale-a & K.Swain

Without Swimming

OPIHI Trajectory

2003–04

Grad students partnered 1:1 with teachers

Species searches with K-12 students

Identified **516 species**; high % invasive and endemic

2005–07

Standardized protocol to collect abundance data

8 schools, 13 sites, 4 islands

First description of community-level patterns in Hawai‘i’s intertidal.

Grad students funded by **NSF GK-12**





Transects laid perpendicular to shore (~3-7) around set place

- Endpoints vary by site, low can vary with tide/waves
- Defined central area; ~3-5 students/transect

Quadrats (⅓ m) along transects—# varies with # students

- **Each quadrat: 25 point-intercepts**
- **Teachers told to aim for 25 quadrats *(cut-point has been 20)***

Island Expansion: Spatial & Temporal Variation

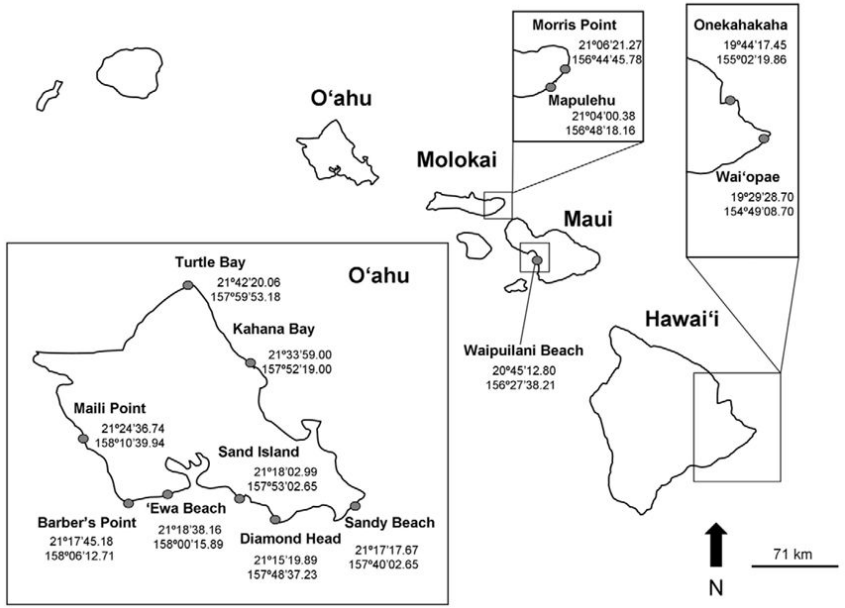
2003–2007

13 sites, 4 islands, 8 schools

First description of community-level patterns at multiple intertidal sites along the Main Hawaiian Islands

- **Spatial variation** both among and within shores
- **Temporal variation** for select sites

Local-scale drivers of structure



Spatial and Temporal Variation in Rocky Intertidal Communities Along the Main Hawaiian Islands

Author(s): Traci Erin Cox , Joanna Philippoff , Erin Baumgartner , Chela J. Zabin , and Celia M. Smith

Validation

Students accurately collect community-level data, but are likely to misidentify cryptic and rare species

Student error rates decreased with teacher field trip experience



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2016–2019

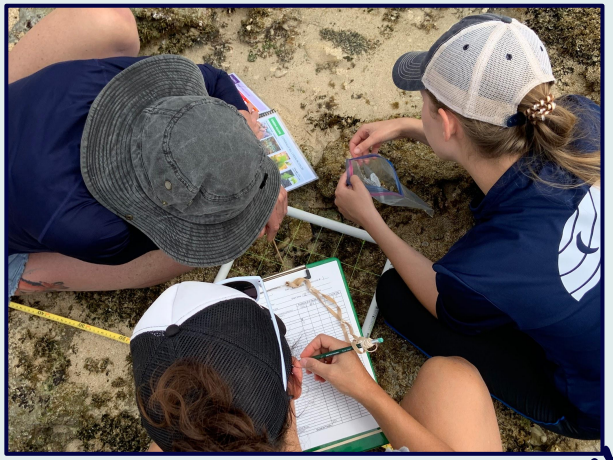
Teacher PD model

Grad students funded by **NSF GK-12**

TEACHER* PROFESSIONAL DEVELOPMENT WORKSHOPS







*Teachers of students aged ~10–18 years

Pictures by J. Philipoff

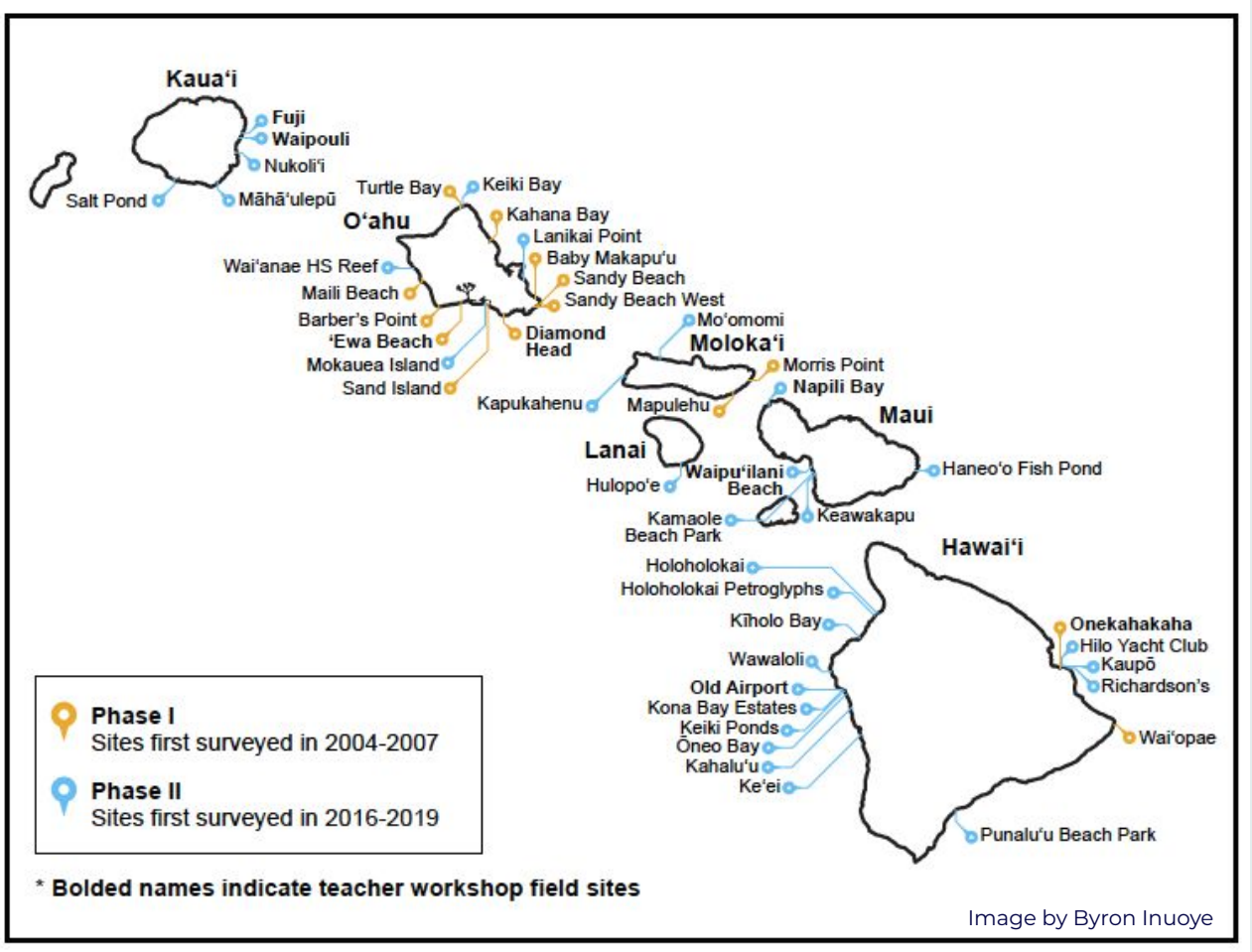
2016–2019:

~90 teachers

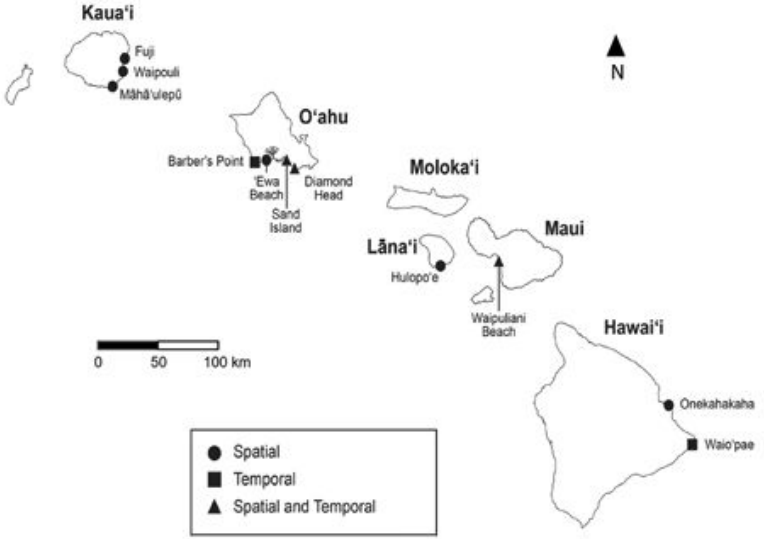
~3,700 students

~180 field trips

46 locations



OPIHI longitudinal data (10 years: 2006/7–2016/17)



Spatial and temporal patterns in Hawai'i's intertidal: decadal changes in benthic community composition

Florybeth Flores La Valle · Jessica Lynn Binder Schaefer · T. Erin Cox · Joanna Philippoff

Spatial (9 sites 5 islands):

- Organisms differ by site, substrate type (basalt vs. limestone), and island

Temporal (5 sites on 3 islands):

- Community structure varies significantly across years and decades.
- Decadal differences were site-specific

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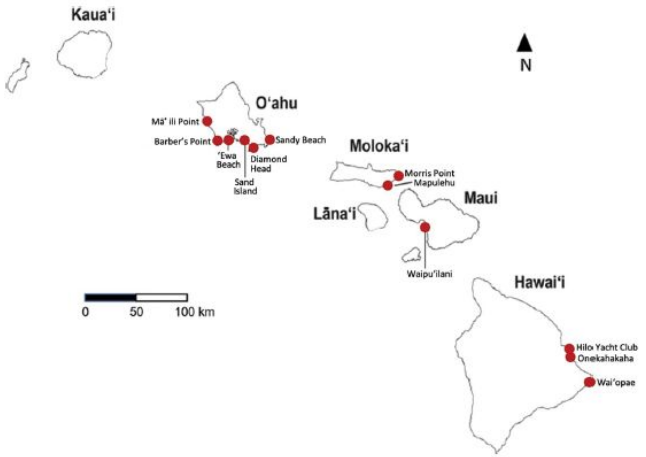
Teacher PD model

90 teachers
3,700 students
46 locations

Demonstrated **change in intertidal ecosystem** over 10 years

NOAA BWET

Intertidal Affected by Sea Surface Temperature (SST)



Spatial variation and antecedent sea surface temperature conditions influence Hawaiian intertidal community structure

Rebecca J. Ward[1,2]*, T. Erin Cox[3], Anuschka Faucci[4], Florybeth Flores La Valle[5], Joanna Philippoff[6], Jessica L. B. Schaefer[7], Ian M. Ware[8], Matthew L. Knope[1,2]

12 OPIHI Intertidal Sites (2004–2019)

- High rates of SST warming (0.40 C) over timeframe

SST Drives Taxonomic & Diversity Shifts

- Diversity increased with rising SST—driven by decrease in turf algae

Critical Role of Antecedent SST (Lagging Effects)

- Temperature conditions 6-12 months prior to sampling strongest drivers for shifting diversity and species abundances

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NOAA BWET

2020–Present

iNaturalist intertidal community survey events in collaboration with **Oceania, UH Sea Grant** and **KUA**

Some 499-collected data (~2020-2025)

2025–2026

National geographic funds O‘ahu & Fiji PD

WHERE YOU COME IN

20+ year of abundance data
Intermittently collected based on funding

Time to look at change over time again!
Important for community (teachers and students—project authenticity)
More interest in Hawaiian intertidal than ever!

Project has been funded on a shoestring

- Education grants
- Most money towards buses, subs, supplies
- Limited GA support

I have limited capacity :(

Side quest—In past abundance studies
Only locations included if sampled 3x a year—what else can do with data?

DATA

Database—included data from 2005 to ~2019

- Downloaded all data before died...Really wild west after that...

Considerations:

- Teachers vary in competence, amount of time can devote to project in classroom
- Different uses (10 year, SST, OPIHI 499) → different inclusion/cleaning protocols

Missing Data:

- Do not have all have data from spring 2026 (but severely disrupted from Kona Lows)
- Some teachers do without support after grant ends—call to teachers for data
- Some not entered (only pictures of data sheets of some 499 years... some data sheets in my office...)

SO MUCH OVERLAP and DUPLICATION

"Cleaned" and "reformatted" for so many things I'm not sure what have and what don't...
I might (?) have more data but need to look in ONE “master” place to check!

Access issues - ASK!! Questions? ASK!!

DATA

Identification level

- Generally to genus except for distinct, dominants, culturally important, invasives
- Reliance on intertidal ID guide

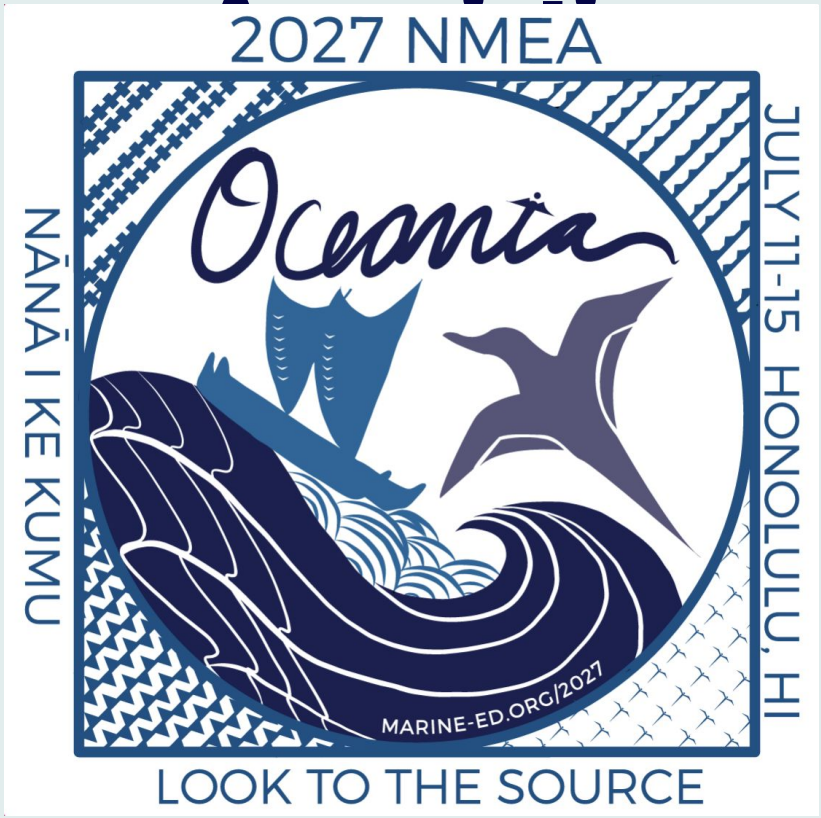
Species names have changed over time

ID resources/materials have become outdated

Use of Hawaiian place names

- Diamond Head = Lēʻahi
- Barbers Point = Kalaeloa
- Sand Island = Māmala Bay = Mauliola

National Marine Educators



Honolulu July 11-15 2027

MAHALO!

QUESTIONS?

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